



M48T559Y

64 Kbit (8Kb x8) TIMEKEEPER[®] SRAM with ADDRESS/DATA MULTIPLEXED

- SOFTWARE and HARDWARE RESET for WATCHDOG TIMER
- REGISTER COMPATIBLE with M48T59 TIMEKEEPER SRAM
- ADDRESS/DATA MULTIPLEXED I/O PINS
- WATCHDOG TIMER - MONITORS OUT of CONTROL PROCESSOR or HUNG BUS
- ALARM with WAKE-UP in BATTERY MODE
- INTEGRATED ULTRA LOW POWER SRAM, REAL TIME CLOCK, POWER-FAIL CONTROL CIRCUIT and BATTERY
- FREQUENCY TEST OUTPUT for REAL TIME CLOCK
- AUTOMATIC POWER-FAIL CHIP DESELECT and WRITE PROTECTION
- WRITE PROTECT VOLTAGE (V_{PFD} = Power-fail Deselect Voltage):
 - M48T559Y: $4.2V \leq V_{PFD} \leq 4.5V$
- PACKAGING INCLUDES a 28-LEAD SOIC and SNAPHAT[®] TOP (to be Ordered Separately)
- SOIC PACKAGE PROVIDES DIRECT CONNECTION for a SNAPHAT TOP CONTAINS the BATTERY and CRYSTAL
- MICROPROCESSOR POWER-ON RESET (Valid even during battery back-up mode)
- PROGRAMMABLE ALARM OUTPUT ACTIVE in the BATTERY BACK-UP MODE

DESCRIPTION

The M48T559Y TIMEKEEPER[®] RAM is an 8K x 8 non-volatile static RAM and real time clock. The monolithic chip is available in the SNAPHAT package to provide a highly integrated battery backed-up memory and real time clock solution.

The 28 pin 330mil SOIC provides sockets with gold plated contacts at both ends for direct connection to a separate SNAPHAT housing containing the battery and crystal. The unique design allows the SNAPHAT battery package to be mounted on top of the SOIC package after the completion of the surface mount process.

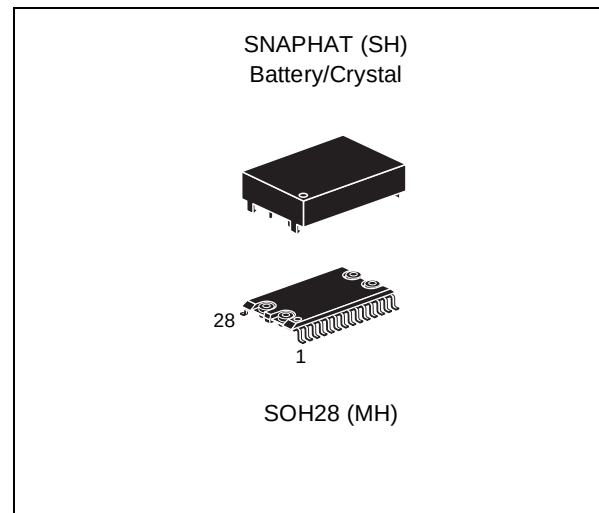


Figure 1. Logic Diagram

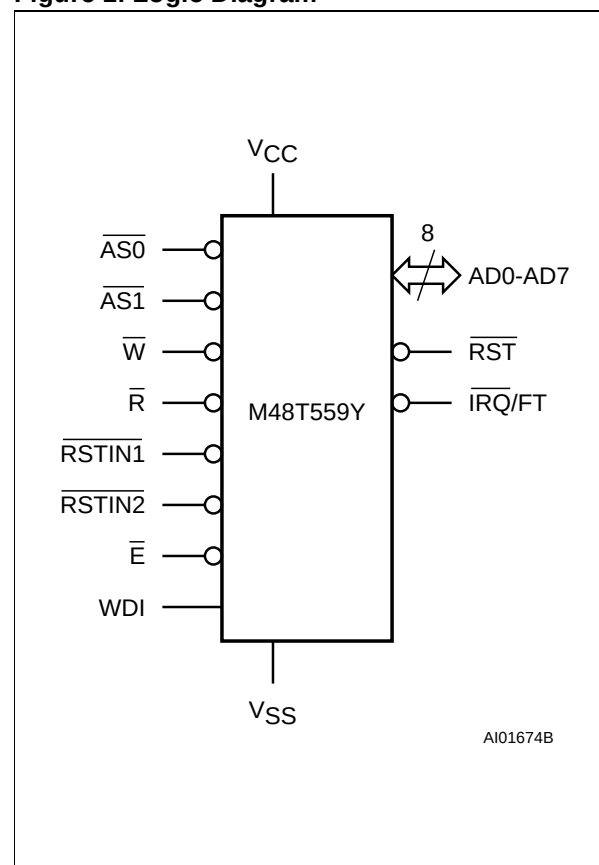


Figure 2. SOIC Connections

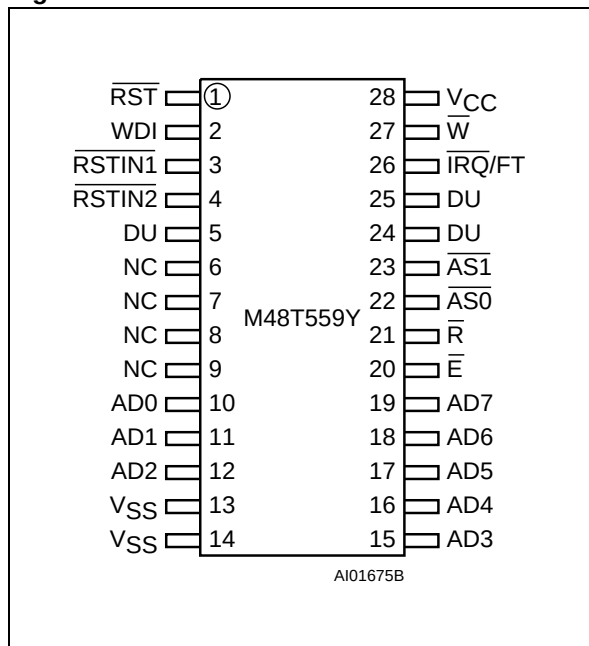


Table 1. Signal Names

AD0-AD7	Address/Data
AS0-AS1	Address Strobes
W	Write Enable
R	Read Enable
E	Chip Enable
WDI	Watchdog Input
RSTIN1-RSTIN2	Reset Input
RST	Power Fail Reset Output (Open Drain)
IRQ/FT	Interrupt / Frequency Test Output (Open Drain)
VCC	Supply Voltage
VSS	Ground
NC	Not Connected Internally
DU	Don't Use must be connected to VCC or VSS

Table 2. Absolute Maximum Ratings

Symbol	Parameter	Value	Unit
T _A	Ambient Operating Temperature	0 to 70	°C
T _{STG}	Storage Temperature (V _{CC} Off, Oscillator Off)	-40 to 85	°C
V _{IO}	Input or Output Voltages	-0.3 to 7	V
V _{CC}	Supply Voltage	-0.3 to 7	V
I _O	Output Current	20	mA
P _D	Power Dissipation	1	W

Note: Stresses greater than those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. This is a stress rating only and functional operation of the device at these or any other conditions above those indicated in the operational section of this specification is not implied. Exposure to the absolute maximum rating conditions for extended periods of time may affect reliability.

CAUTION: Negative undershoots below -0.3V are not allowed on any pin while in the Battery Back-up mode.

CAUTION: Do NOT wave solder SOIC to avoid damaging SNAPHAT sockets.

Insertion of the SNAPHAT housing after reflow prevents potential battery and crystal damage due to the high temperatures required for device surface-mounting. The SNAPHAT housing is keyed to prevent reverse insertion.

The SOIC and battery/crystal packages are shipped separately in plastic anti-static tubes or in Tape & Reel form. For the 28 lead SOIC, the battery/crystal package (i.e. SNAPHAT) part number is "M4T28-BR12SH1".

Caution: Do not place the SNAPHAT battery/crystal top in conductive foam, as this will drain the lithium button-cell battery.

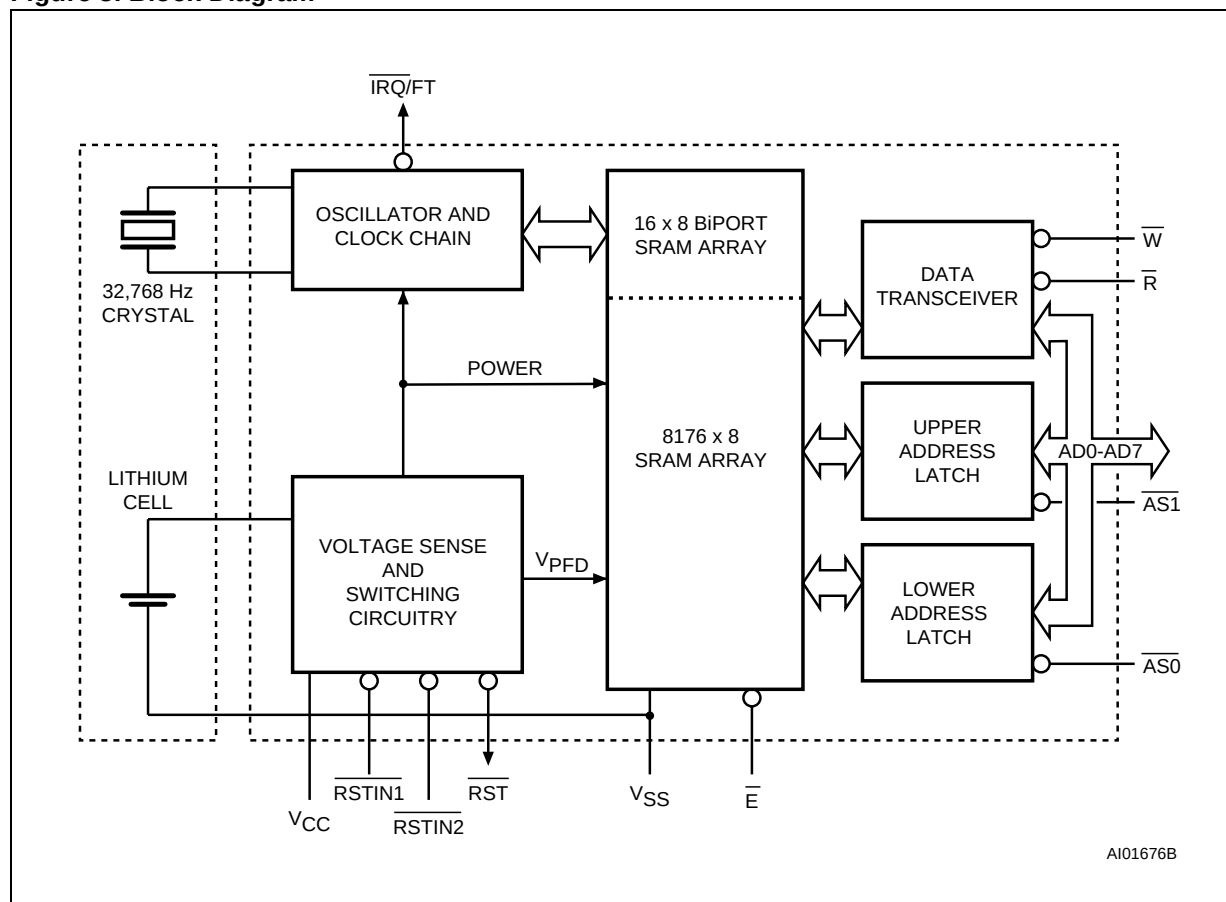
As Figure 3 shows, the static memory array and the quartz controlled clock oscillator of the M48T559Y are integrated on one silicon chip. The

two circuits are interconnected at the upper eight memory locations to provide user accessible BYTEWIDE™ clock information in the bytes with addresses 1FF8h-1FFFh. The clock locations contain the year, month, date, day, hour, minute, and second in 24 hour BCD format. Corrections for 28, 29 (leap year), 30, and 31 day months are made automatically. Byte 1FF8h is the clock control register. This byte controls user access to the clock information and also stores the clock calibration setting.

The eight clock bytes are not the actual clock counters themselves; they are memory locations consisting of BiPORT™ read/write memory cells.

The M48T559Y includes a clock control circuit which updates the clock bytes with current infor-

Figure 3. Block Diagram

Table 3. Operating Modes ⁽¹⁾

Mode	V _{CC}	$\bar{E}$	$\bar{R}$	$\bar{W}$	AD0-AD7	Power
Deselect	4.5V to 5.5V	V _{IH}	X	X	High Z	Standby ⁽³⁾
Write		V _{IL}	V _{IH}	V _{IL}	D _{IN}	Active
Read		V _{IL}	V _{IL}	V _{IH}	D _{OUT}	Active
Read		V _{IL}	V _{IH}	V _{IH}	High Z	Active
Deselect	V _{SO} to V _{PFD} (min) ⁽²⁾	X	X	X	High Z	CMOS Standby
Deselect	≤ V _{SO}	X	X	X	High Z	Battery Back-up Mode

Note: 1. X = V_{IH} or V_{IL}; V_{SO} = Battery Back-up Switchover Voltage.

2. See Table 7 for details.

3. AD0-AD7, $\bar{AS0}$, $\bar{AS1}$ active when $\bar{E}$ is high and V_{CC} > V_{PFD}.

information once per second. The information can be accessed by the user in the same manner as any other location in the static memory array.

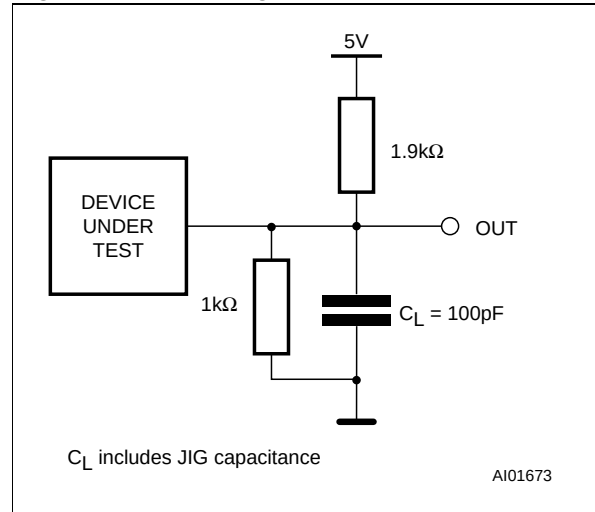
The M48T559Y also has its own Power-fail Detect circuit. The control circuitry constantly monitors the single 5V supply for an out of tolerance condition.

When V_{CC} is out of tolerance, the circuit write protects the SRAM, providing a high degree of data security in the midst of unpredictable system operation brought on by low V_{CC}. As V_{CC} falls below approximately 3V, the control circuitry connects the battery which maintains data and clock operation until valid power returns.

Table 4. AC Measurement Conditions

Input Rise and Fall Times	$\leq 5\text{ns}$
Input Pulse Voltages	0 to 3V
Input and Output Timing Ref. Voltages	1.5V

Note that Output Hi-Z is defined as the point where data is no longer driven.

Figure 4. AC Testing Load Circuit**Table 5. Capacitance (1, 2)**

($T_A = 25^\circ\text{C}$, $f = 1\text{ MHz}$)

Symbol	Parameter	Test Condition	Min	Max	Unit
C_{IN}	Input Capacitance	$V_{IN} = 0\text{V}$		10	pF
$C_{IO}^{(3)}$	Input / Output Capacitance	$V_{OUT} = 0\text{V}$		10	pF

Note: 1. Effective capacitance measured with power supply at 5V.
 2. Sampled only, not 100% tested.
 3. Outputs deselected.

Table 6. DC Characteristics

($T_A = 0$ to 70°C ; $V_{CC} = 4.5\text{V}$ to 5.5V)

Symbol	Parameter	Test Condition	Min	Max	Unit
$I_{LI}^{(1)}$	Input Leakage Current	$0\text{V} \leq V_{IN} \leq V_{CC}$		± 1	μA
$I_{LO}^{(1)}$	Output Leakage Current	$0\text{V} \leq V_{OUT} \leq V_{CC}$		± 5	μA
$I_{LRST}^{(2)}$	Input Leakage Current	$0\text{V} \leq V_{IN} \leq V_{CC}$		100	μA
I_{CC}	Supply Current	Outputs open		50	mA
I_{CC1}	Supply Current (Standby) TTL	$\bar{E} = V_{IH}$		10	mA
$I_{CC2}^{(3)}$	Supply Current (Standby) CMOS	$\bar{E} = V_{CC} - 0.2\text{V}$		7	mA
$V_{IL}^{(4)}$	Input Low Voltage		-0.3	0.8	V
V_{IH}	Input High Voltage		2.2	$V_{CC} + 0.3$	V
V_{OL}	Output Low Voltage	$I_{OL} = 2.1\text{mA}$		0.4	V
	Output Low Voltage ($\overline{IRQ}/\text{FT}$) ⁽⁵⁾	$I_{OL} = 10\text{mA}$		0.4	V
$V_{OH}^{(6)}$	Output High Voltage	$I_{OH} = -1\text{mA}$	2.4		V

Note: 1. Outputs deselected.
 2. Input leakage current on input RESET pins.
 3. AD0-AD7, $\overline{AS0}$, $\overline{AS1}$ active when $\bar{E}$ is high and $V_{CC} > V_{PFD}$.
 4. Negative spikes of -1V allowed for up to 10ns once per cycle.
 5. The $\overline{IRQ}$ pins is Open Drain.
 6. Measured with Control Bits set as follows: R = '1'; W, ST, FT = '0'.

Table 7. Power Down/Up Trip Points DC Characteristics ⁽¹⁾
 $(T_A = 0 \text{ to } 70^\circ\text{C})$

Symbol	Parameter	Min	Typ	Max	Unit
V_{PFD}	Power-fail Deselect Voltage	4.2	4.35	4.5	V
V_{SO}	Battery Back-up Switchover Voltage		3.0		V
$t_{DR}^{(2)}$	Expected Data Retention Time	7			YEARS

Note: 1. All voltages referenced to V_{SS} .

2. At 25°C .

Table 8. Power Down/Up AC Characteristics
 $(T_A = 0 \text{ to } 70^\circ\text{C})$

Symbol	Parameter	Min	Max	Unit
t_{PD}	$\overline{E}$ at V_{IH} before Power Down	0		μs
$t_F^{(1)}$	$V_{PFD}(\text{max})$ to $V_{PFD}(\text{min})$ V_{CC} Fall Time	300		μs
$t_{FB}^{(2)}$	$V_{PFD}(\text{min})$ to V_{SS} V_{CC} Fall Time	10		μs
t_R	$V_{PFD}(\text{min})$ to $V_{PFD}(\text{max})$ V_{CC} Rise Time	10		μs
t_{RB}	V_{SS} to $V_{PFD}(\text{min})$ V_{CC} Rise Time	1		μs
t_{REC}	$V_{PFD}(\text{max})$ to $\overline{RST}$ High	40	200	ms

Note: 1. $V_{PFD}(\text{max})$ to $V_{PFD}(\text{min})$ fall time of less than t_F may result in deselection/write protection not occurring until $200\mu\text{s}$ after V_{CC} passes $V_{PFD}(\text{min})$.

2. $V_{PFD}(\text{min})$ to V_{SS} fall time of less than t_{FB} may cause corruption of RAM data.

Figure 5. Power Down/Up Mode AC Waveforms

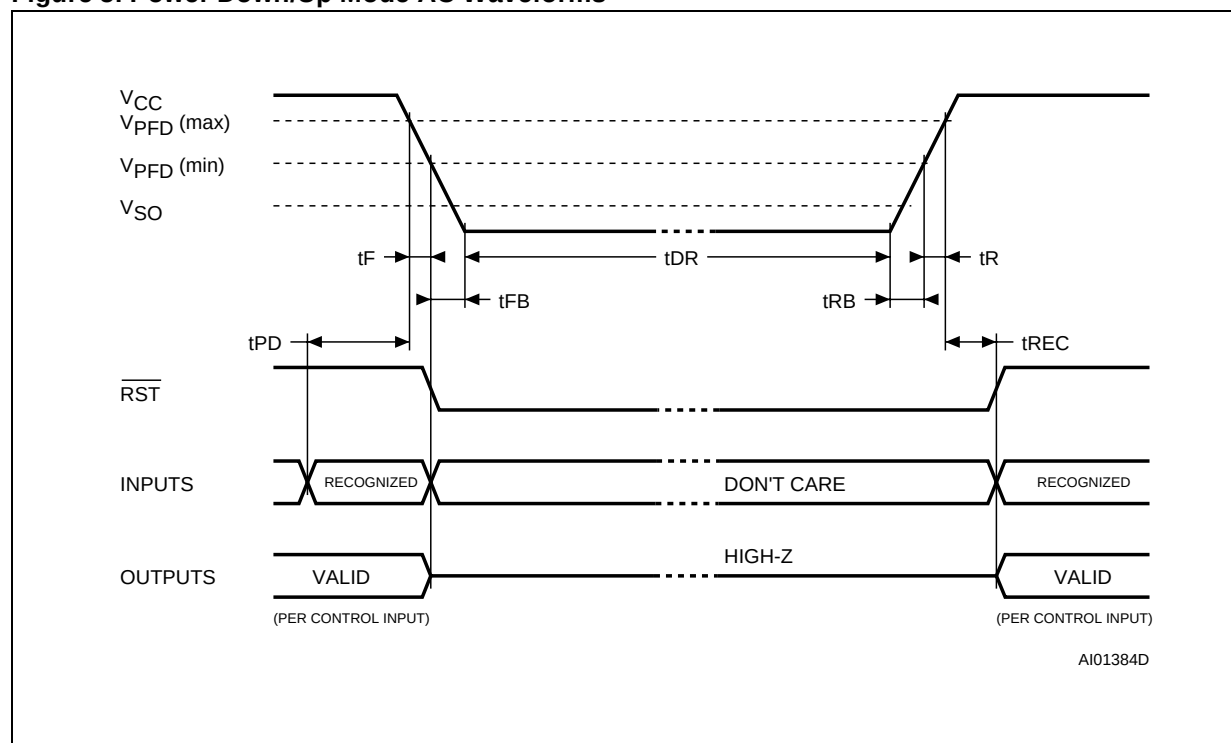


Table 9. AC Characteristics(T_A = 0 to 70 °C; V_{CC} = 4.5V to 5.5V)

Symbol	Parameter	M48T559Y		Unit
		Min	Max	
t _{AS}	Address Setup Time	20		ns
t _{AH}	Address Hold Time	0		ns
t _{DS}	Data Setup Time	60		ns
t _{DH}	Data Hold Time	0		ns
t _{R_{LDV}}	Read Enable Access Time		70	ns
t _{R_{LRH}}	$\overline{R}$ Pulse Width Low	70		ns
t _{R_{HDZ}}	Read Enable High to Output High Z		25	ns
t _{WLWH}	$\overline{W}$ Pulse Width Low	50		ns
t _{E_{LEH}}	$\overline{E}$ Pulse Width Low	50		ns
t _{ASLASH}	$\overline{AS0}$, $\overline{AS1}$ Pulse Width Low	15		ns
t _{ASHRL}	$\overline{AS0}$, $\overline{AS1}$ High to $\overline{R}$ Low	15		ns
t _{ASHWL}	$\overline{AS0}$, $\overline{AS1}$ High to $\overline{W}$ Low	15		ns
t _{ELRL}	Chip Enable Low to Read Enable Low	0		ns
t _{E_{HDZ}}	Chip Enable High to Data Output Hi-Z	0		ns
t _{ELWL}	Chip Enable Low to Write Enable Low	0		ns

RAM OPERATION

Four control signals, $\overline{AS0}$, $\overline{AS1}$, $\overline{R}$ and $\overline{W}$, are used to access the M48T559Y. The address latches are loaded from the address/data bus in response to rising edge signals applied to the Address Strobe 0 ($\overline{AS0}$) and Address Strobe 1 ($\overline{AS1}$) signals. $\overline{AS0}$ is used to latch the lower 8 bits of address, and $\overline{AS1}$ is used to latch the upper 5 bits of address.

It is not however necessary to follow any particular order. The inputs are in parallel for the two address bytes (upper and lower) and can be latched in any order as long as the correct strobe is used. It is necessary to meet the set-up and hold times given in the AC specifications with valid address information in order to properly latch the address. If the upper and/or lower order addresses are correct from a prior cycle, it is not necessary to repeat the address latching sequence.

A write operation requires valid data to be placed on the bus (AD0-AD7), followed by the activation of the Write Enable ($\overline{W}$) line. Data on the bus will be written to the RAM, provided that the write timing specifications are met. During a read cycle, the Read Enable ($\overline{R}$) signal is driven active. Data from the RAM will become valid on the bus provided that the RAM read access timing specifications are met.

The $\overline{W}$ and $\overline{R}$ signals should never be active at the same time. In addition, $\overline{E}$ must be active before any control line is recognized (except for AD0-AD7 and $\overline{AS0}$, $\overline{AS1}$).

RESET INPUT

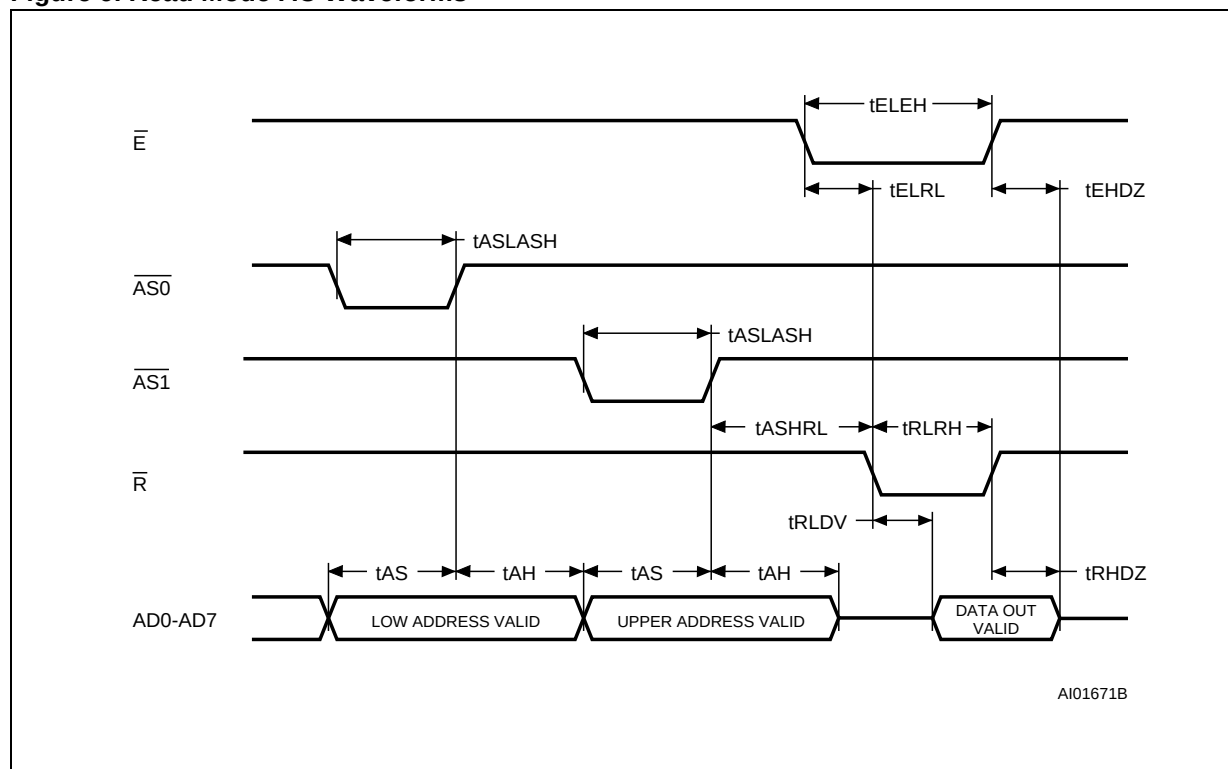
The M48T559Y provides two debounced inputs which can generate an output Reset. The duration and function of the Reset output is identical to a Reset generated by a power cycle. Pulses shorter than t_{R1} and t_{R2} will not generate a Reset condition (see Table 12 and Figure 13).

DATA RETENTION MODE

Should the supply voltage decay, the RAM will automatically power-fail deselect, write protecting itself when V_{CC} falls within the V_{PFD} (max), V_{PFD} (min) window. All outputs become high impedance, and all inputs are treated as "don't care."

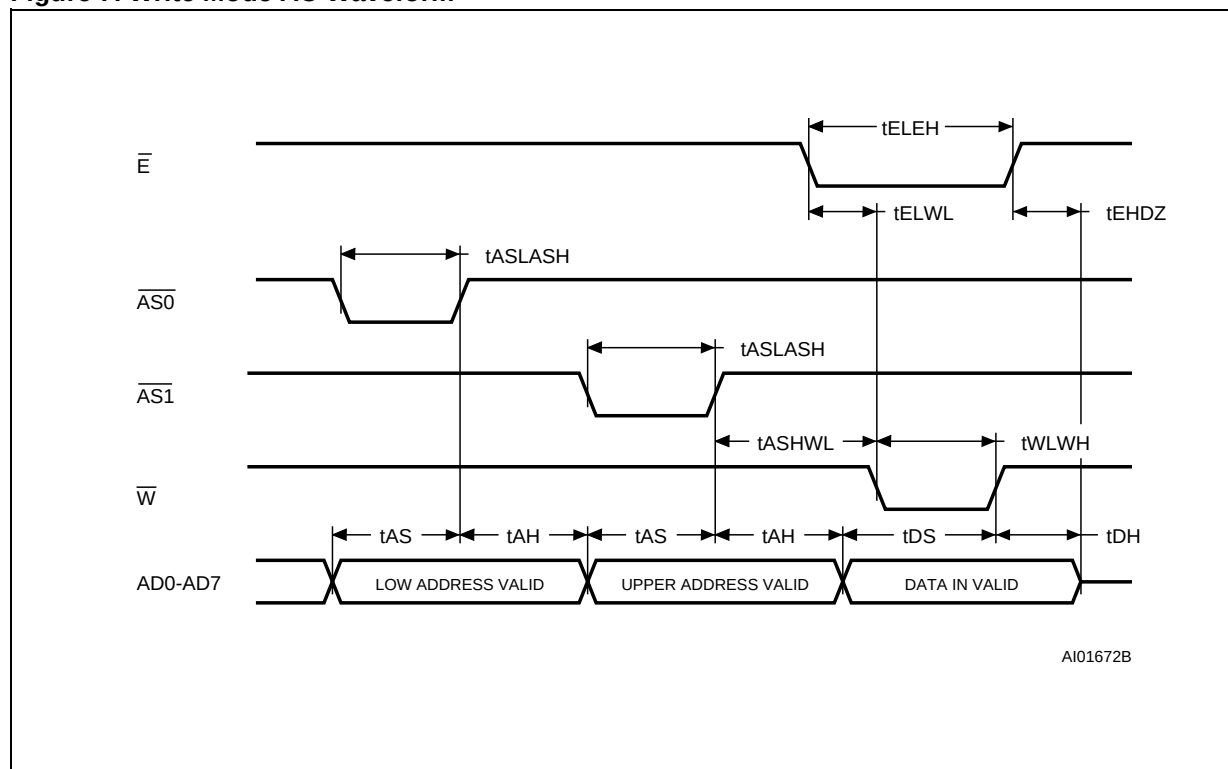
Note: A power failure during a write cycle may corrupt data at the currently addressed location, but does not jeopardize the rest of the RAM's content. At voltages below V_{PFD} (min), the user can be assured the memory will be in a write protected state, provided the V_{CC} fall time is not less than t_F.

Figure 6. Read Mode AC Waveforms



Note: AD5-AD7 are don't care when latching upper address.

Figure 7. Write Mode AC Waveform



Note: AD5-AD7 are don't care when latching upper address.

Table 10. Register Map

Address	Data								Function/Range BCD Format	
	D7	D6	D5	D4	D3	D2	D1	D0		
1FFFh	10 Years				Year				Year	00-99
1FFEh	0	0	0	10 M	Month				Month	01-12
1FFDh	0	0	10 Date		Date				Date	01-31
1FFCh	0	FT	0	0	0	Day			Day	01-07
1FFBh	0	0	10 Hours		Hours				Hour	00-23
1FFAh	0	10 Minutes			Minutes				Minutes	00-59
1FF9h	ST	10 Seconds			Seconds				Seconds	00-59
1FF8h	W	R	S	Calibration					Control	
1FF7h	WDS	BMB4	BMB3	BMB2	BMB1	BMB0	RB1	RB0	Watchdog	
1FF6h	AFE	Y	ABE	Y	Y	Y	Y	Y	Interrupts	
1FF5h	RPT4	Y	Al. 10 Date		Alarm Date				Alarm Date	01-31
1FF4h	RPT3	Y	Al. 10 Hours		Alarm Hours				Alarm Hours	00-23
1FF3h	RPT2	Alarm 10 Minutes			Alarm Minutes				Alarm Minutes	00-59
1FF2h	RPT1	Alarm 10 Seconds			Alarm Seconds				Alarm Seconds	00-59
1FF1h	Y	Y	Y	Y	Y	Y	Y	Y	Unused	
1FF0h	WDF	AF	Z	BL	Z	Z	Z	Z	Flags	

Keys: S = SIGN Bit
 FT = FREQUENCY TEST Bit
 R = READ Bit
 W = WRITE Bit
 ST = STOP Bit
 0 = Must be set to zero
 Y = '1' or '0'
 Z = '0' and are Read only
 AF = Alarm Flag

BL = Battery Low
 WDS = Watchdog Steering Bit
 BMB0-BMB4 = Watchdog Multiplier Bits
 RB0-RB1 = Watchdog Resolution Bits
 AFE = Alarm Flag Enable
 ABE = Alarm in Battery Back-up Mode Enable
 RPT1-RPT4 = Alarm Repeat Mode Bits
 WDF = Watchdog Flag

The M48T559Y may respond to transient noise spikes on V_{CC} that reach into the deselect window during the time the device is sampling V_{CC} . Therefore, decoupling of the power supply lines is recommended.

When V_{CC} drops below V_{SO} , the control circuit switches power to the internal battery which preserves data and powers the clock. The internal button cell will maintain data in the M48T559Y for an accumulated period of at least 7 years when V_{CC} is less than V_{SO} . As system power returns and V_{CC} rises above V_{SO} , the battery is disconnected, and the power supply is switched to exter-

nal V_{CC} . Write protection continues until V_{CC} reaches V_{PFD} (max) plus t_{REC} . For more information on Battery Storage Life refer to the Application Note AN1012.

POWER-ON RESET

The M48T559Y continuously monitors V_{CC} . When V_{CC} falls to the power fail detect trip point, the $\overline{RST}$ pulls low (open drain) and remains low on power-up for 40ms to 200ms after V_{CC} passes V_{PFD} . An external pull-up resistor to V_{CC} is required (1k Ω resistor is recommended). The reset pulse remains active with V_{CC} at V_{SS} .

CLOCK OPERATIONS

Reading the Clock

Updates to the TIMEKEEPER registers should be halted before clock data is read to prevent reading data in transition. Because the BiPORT TIMEKEEPER cells in the RAM array are only data registers, and not the actual clock counters, updating the registers can be halted without disturbing the clock itself.

Updating is halted when a '1' is written to the READ bit, D6 in the Control register (1FF8h). As long as a '1' remains in that position, updating is halted.

After a halt is issued, the registers reflect the count; that is, the day, date, and the time that were current at the moment the halt command was issued.

All of the TIMEKEEPER registers are updated simultaneously. A halt will not interrupt an update in progress. Updating is within a second after the bit is reset to a '0'.

Setting the Clock

Bit D7 of the Control register (1FF8h) is the WRITE bit. Setting the WRITE bit to a '1', like the READ bit, halts updates to the TIMEKEEPER registers. The user can then load them with the correct day, date, and time data in 24 hour BCD format (see Table 9). Resetting the WRITE bit to a '0' then transfers the values of all time registers (1FF9h-1FFFh) to the actual TIMEKEEPER counters and allows normal operation to resume. After the WRITE bit is reset, the next clock update will occur in one second.

See the Application Note AN923 "TIMEKEEPER rolling into the 21st century" for information on Century Rollover.

Stopping and Starting the Oscillator

The oscillator may be stopped at any time. If the device is going to spend a significant amount of time on the shelf, the oscillator can be turned off to minimize current drain on the battery. The STOP bit is the MSB of the seconds register. Setting it to a '1' stops the oscillator. The M48T559Y is shipped from STMicroelectronics with the STOP bit set to a '1'. When reset to a '0', the M48T559Y oscillator starts within one second.

Calibrating the Clock

The M48T559Y is driven by a quartz controlled oscillator with a nominal frequency of 32,768Hz. The devices are tested not to exceed 35 ppm (parts per million) oscillator frequency error at 25°C, which equates to about ± 1.53 minutes per month. With the calibration bits properly set, the accuracy of each M48T559Y improves to better than ± 4 ppm at 25°C.

The oscillation rate of any crystal changes with temperature (see Figure 10). Most clock chips compensate for crystal frequency and temperature shift error with cumbersome trim capacitors. The M48T559Y design, however, employs periodic counter correction. The calibration circuit adds or subtracts counts from the oscillator divider circuit at the divide by 256 stage, as shown in Figure 9. The number of times pulses are blanked (subtracted, negative calibration) or split (added, positive calibration) depends upon the value loaded into the five bit Calibration byte found in the Control Register. Adding counts speeds the clock up, subtracting counts slows the clock down.

Figure 8. Clock Calibration

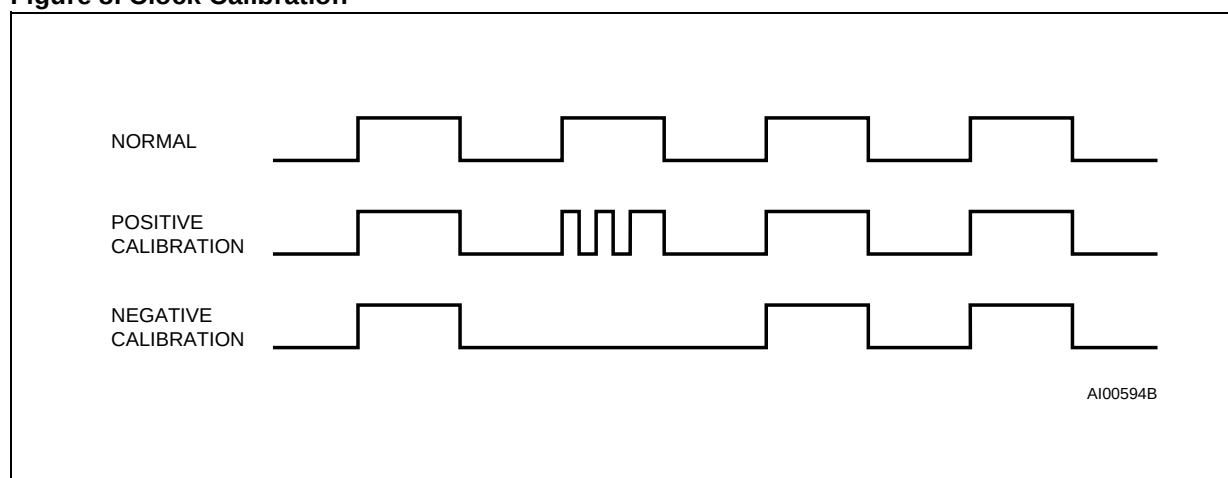
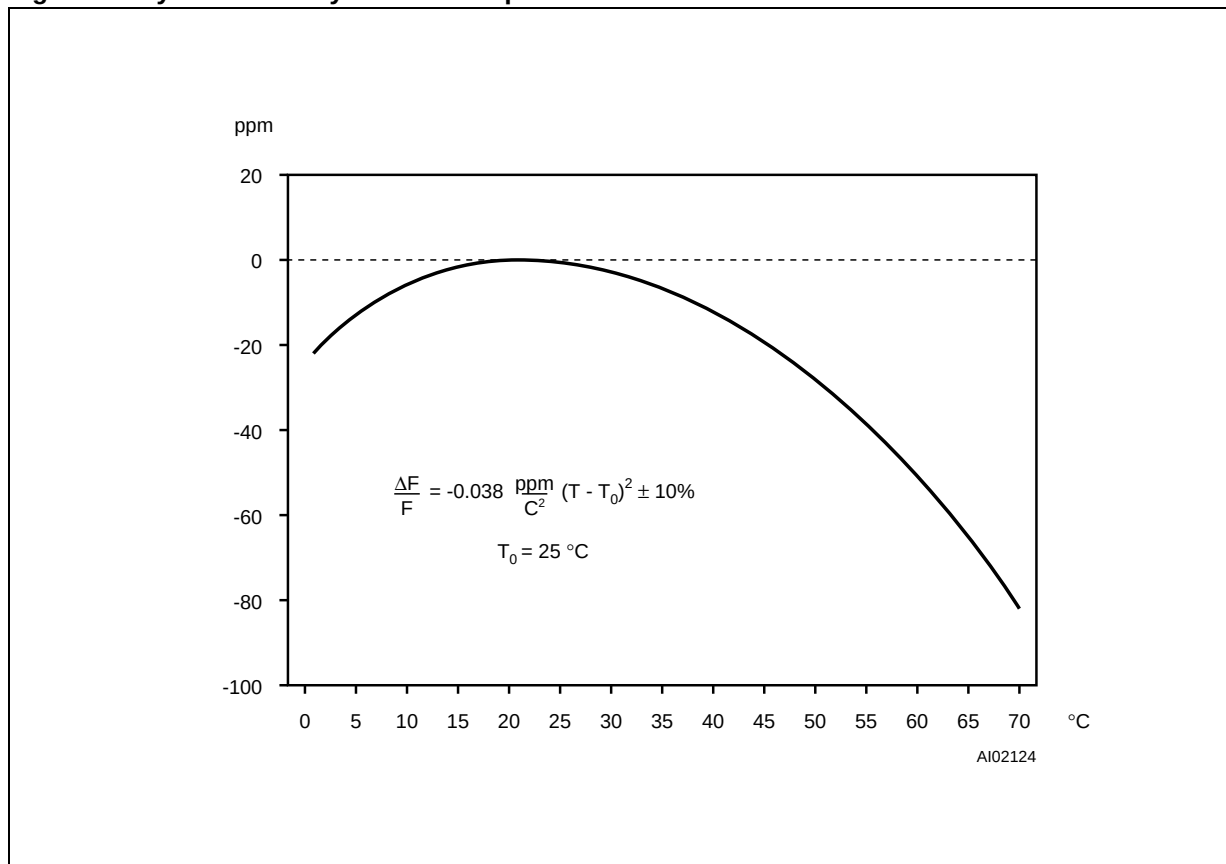


Figure 9. Crystal Accuracy Across Temperature



The Calibration byte occupies the five lower order bits (D4-D0) in the Control register (1FF8h). These bits can be set to represent any value between 0 and 31 in binary form. Bit D5 is a Sign bit; '1' indicates positive calibration, '0' indicates negative calibration. Calibration occurs within a 64 minute cycle. The first 62 minutes in the cycle may, once per minute, have one second either shortened by 128 or lengthened by 256 oscillator cycles.

If a binary '1' is loaded into the register, only the first 2 minutes in the 64 minute cycle will be modified; if a binary 6 is loaded, the first 12 will be affected, and so on.

Therefore, each calibration step has the effect of adding 512 or subtracting 256 oscillator cycles for every 125,829,120 actual oscillator cycles, that is, +4.068 or -2.034 ppm of adjustment per calibration step in the calibration register. Assuming that the oscillator is in fact running at exactly 32,768Hz, each of the 31 increments in the Calibration byte would represent +10.7 or -5.35 seconds per month which corresponds to a total range of +5.5 or - 2.75 minutes per month.

Two methods are available for ascertaining how much calibration a given M48T559Y may require.

The first involves simply setting the clock, letting it run for a month and comparing it to a known accurate reference (like WWV broadcasts). While that may seem crude, it allows the designer to give the end user the ability to calibrate his clock as his environment may require, even after the final product is packaged in a non-user serviceable enclosure. All the designer has to do is provide a simple utility that accesses the Calibration byte.

The second approach is better suited to a manufacturing environment, and involves the use of the IRQ/FT pin. The pin will toggle at 512Hz when the Stop bit (D7 of 1FF9h) is '0', the FT bit (D6 of 1FFCh) is '1', the AFE bit (D7 of 1FF6h) is '0', and the Watchdog Steering bit (D7 of 1FF7h) is '1' or the Watchdog Register is reset (1FF7h = 0).

Any deviation from 512Hz indicates the degree and direction of oscillator frequency shift at the test temperature. For example, a reading of 512.01024 Hz would indicate a +20 ppm oscillator frequency error, requiring a -10 (WR001010) to be loaded into the Calibration Byte for correction. Note that setting or changing the Calibration Byte does not affect the Frequency test output frequency.

The FT bit is cleared on power-down.

For more information on calibration, see the Application Note AN934 "TIMEKEEPER Calibration".

The $\overline{\text{IRQ/FT}}$ pin is an open drain output which requires a pull-up resistor for proper operation. A 500-10k Ω resistor is recommended in order to control the rise time.

SETTING ALARM CLOCK

Registers 1FF5h-1FF2h contain the alarm settings. The alarm can be configured to go off at a prescribed time on a specific day of the month or repeat every day, hour, minute, or second. It can also be programmed to go off while the M48T559Y is in the battery back-up mode of operation to serve as a system wake-up call.

RPT1-RPT4 put the alarm in the repeat mode of operation. Table 11 shows the possible configurations. Codes not listed in the table default to the once per second mode to quickly alert the user of an incorrect alarm setting.

Note: User must transition address (or toggle chip enable) to see Flag bit change.

When the clock information matches the alarm clock settings based on the match criteria defined by RPT1-RPT4, AF (Alarm Flag) is set. If AFE (Alarm Flag Enable) is also set, the alarm condition activates the $\overline{\text{IRQ/FT}}$ pin. To disable alarm, write '0' to the Alarm Date registers and RPT1-4. The alarm flag and the $\overline{\text{IRQ/FT}}$ output are cleared by a read to the Flags register as shown in Figure 11.

Note: If an alarm condition occurs while the flags register address is latched into the address buffer, the alarm flag will not be set until an address other than the flags register (1FF0h) is latched into the address buffer. This will insure that the alarm flag will not be inadvertently reset while reading the flag register. To properly check to see if an alarm condition has occurred while reading the flag register, the user is required to latch, read or write to an alternate address and then re-read the alarm flag.

The $\overline{\text{IRQ/FT}}$ pin can also be activated in the battery back-up mode. The $\overline{\text{IRQ/FT}}$ will go low if an alarm occurs and both ABE (Alarm in Battery Back-up Mode Enable) and AFE are set. The ABE and AFE bits are reset during power-up, therefore an alarm generated during power-up will only set AF. The user can read the Flag Register at system boot-up to determine if an alarm was generated while the M48T559Y was in the deselect mode during power-up. Figure 12 illustrates the back-up mode alarm timing.

Table 11. Alarm Repeat Mode

RPT4	RPT3	RPT2	RPT1	Alarm Activated
1	1	1	1	Once per Second
1	1	1	0	Once per Minute
1	1	0	0	Once per Hour
1	0	0	0	Once per Day
0	0	0	0	Once per Month

Figure 10. Interrupt Reset Waveforms

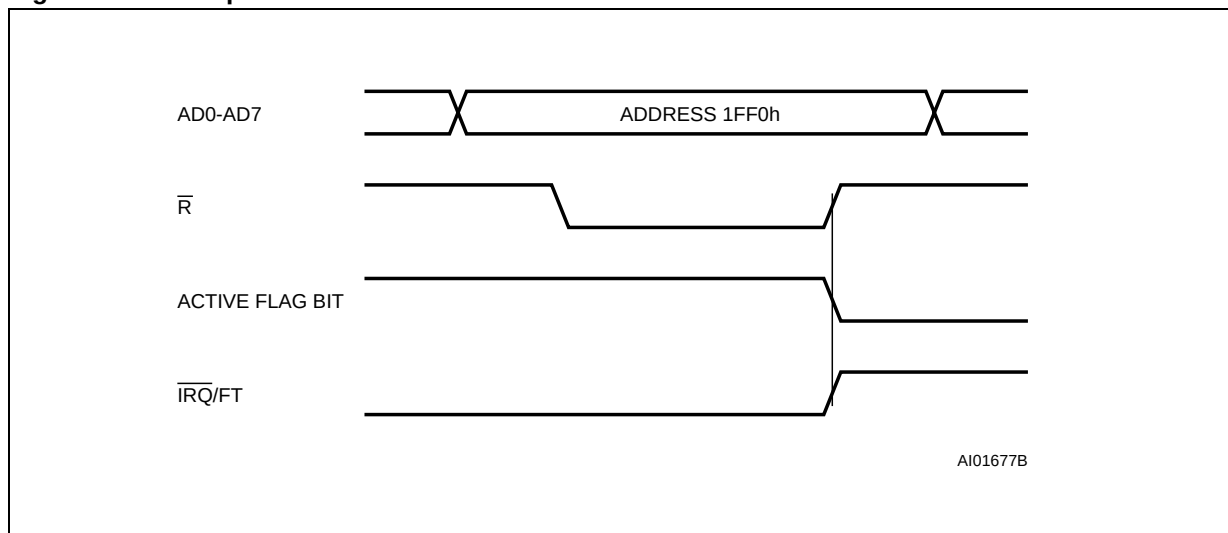
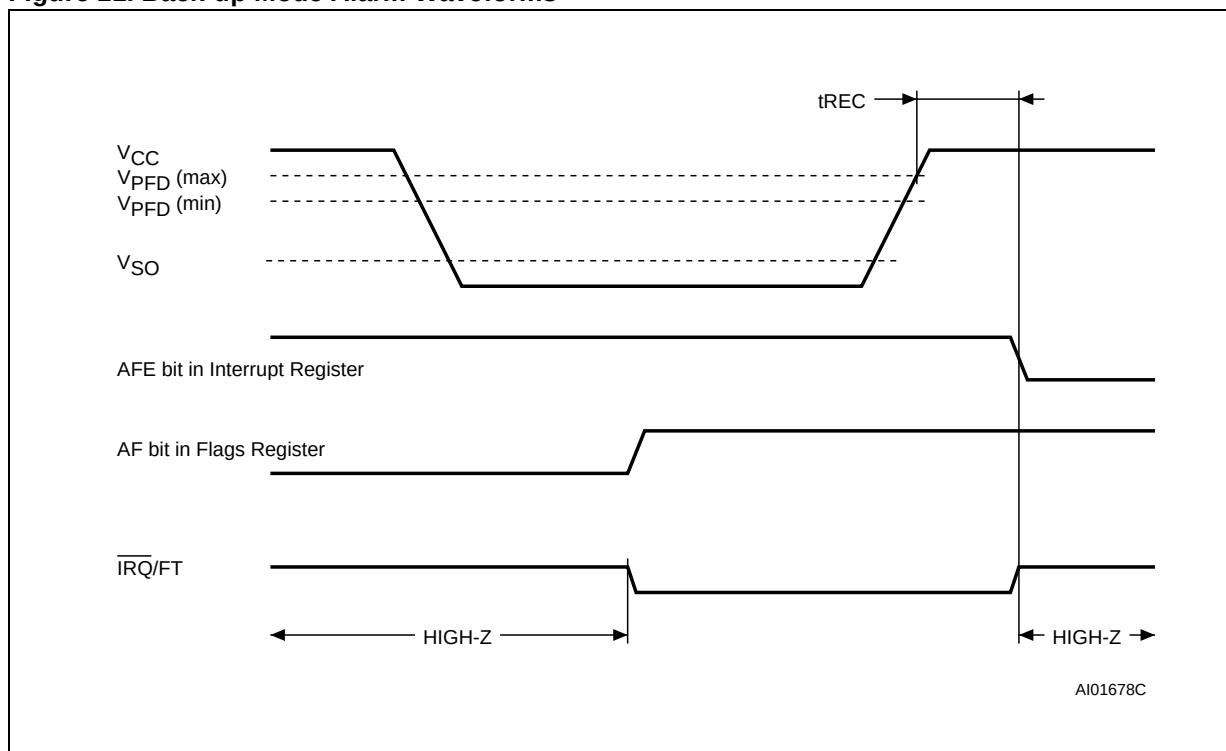


Figure 11. Back-up Mode Alarm Waveforms



WATCHDOG TIMER

The watchdog timer can be used to detect an out-of-control microprocessor. The user programs the watchdog timer by setting the desired amount of time-out into the eight bit Watchdog Register (Address 1FF7h). The five bits (BMB4-BMB0) store a binary multiplier and the two lower order bits (RB1-RB0) select the resolution, where 00 = 1/16 second, 01 = 1/4 second, 10 = 1 second, and 11 = 4 seconds. The amount of time-out is then determined to be the multiplication of the five bit multiplier value with the resolution. (For example: writing 00001110 in the Watchdog Register = 3 x 1 or 3 seconds).

Note: Accuracy of timer is within $\pm$ the selected resolution.

If the processor does not reset the timer within the specified period, the M48T559Y sets the WDF (Watchdog Flag) and generates a watchdog interrupt or a microprocessor reset. WDF is reset by reading the Flags Register (Address 1FF0h).

The most significant bit of the Watchdog Register is the Watchdog Steering Bit. When set to a '0', the watchdog will activate the $\overline{IRQ}/FT$ pin when timed-out. When WDS is set to a '1', the watchdog will output a negative pulse on the $\overline{RST}$ pin for a duration of 40ms to 200ms. The Watchdog register will reset to a '0' at the end of a watchdog time-out when the WDS bit is set to a '1'.

The watchdog timer can be reset by two methods:

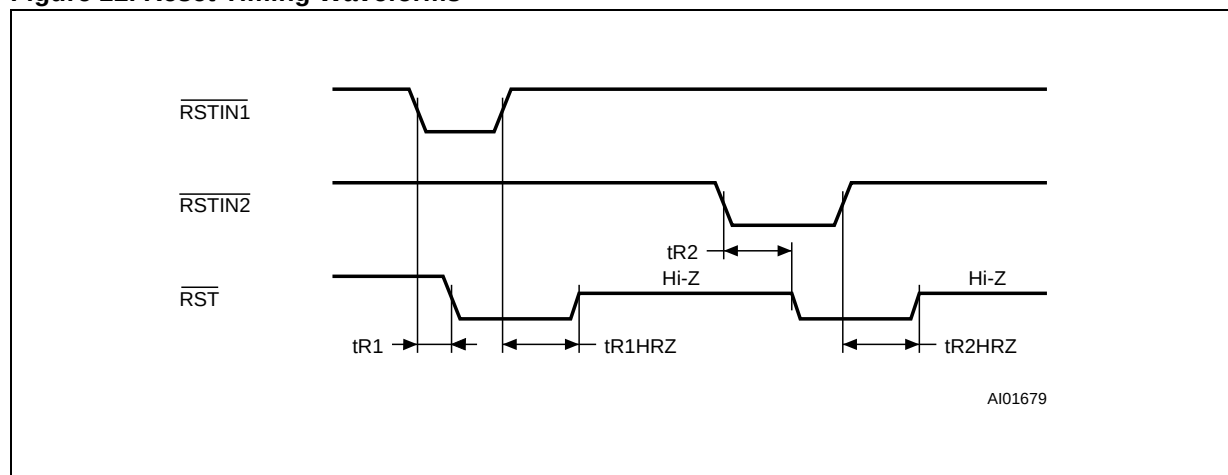
- a transition (high-to-low or low-to-high) can be applied to the Watchdog input pin (WDI)
- or
- the microprocessor can perform a write of the Watchdog Register. The time-out period then starts over. The WDI pin contains a pull-up resistor and therefore can be left unconnected if not used.

The watchdog timer will be reset on each transition (edge) seen by the WDI pin. In order to perform a software reset of the Watchdog timer, the original time-out period can be written into the Watchdog Register, effectively restarting the count-down cycle.

Should the watchdog timer time-out, and the WDS bit is programmed to output an interrupt, a value of 00h needs to be written to the Watchdog Register in order to clear the $\overline{IRQ}/FT$ pin. This will also disable the watchdog function until it is again programmed correctly. A read of the Flags Register will reset the Watchdog Flag (D7; Register MSB-15).

The watchdog function is automatically disabled upon power-down and the Watchdog Register is cleared. If the watchdog function is set to output to the $\overline{IRQ}/FT$ pin and the frequency test function is activated, the watchdog or alarm function prevails and the frequency test function is denied.

Figure 12. Reset Timing Waveforms

Table 12. Reset AC Characteristics ($T_A = 0$ to 70°C ; $V_{CC} = 4.5\text{V}$ to 5.5V)

Symbol	Parameter	Min	Max	Unit
t_{R1}	$\overline{\text{RSTIN1}}$ Low to $\overline{\text{RST}}$ Low	50	200	ns
t_{R2}	$\overline{\text{RSTIN2}}$ Low to $\overline{\text{RST}}$ Low	20	100	ms
t_{R1HRZ}	$\overline{\text{RSTIN1}}$ High to $\overline{\text{RST}}$ Hi-Z	40	200	ms
t_{R2HRZ}	$\overline{\text{RSTIN2}}$ High to $\overline{\text{RST}}$ Hi-Z	40	200	ms

BATTERY LOW WARNING

The M48T559Y checks its battery voltage on power-up. The BL (Battery Low) bit (D4 of 1FF0h) will be set on power-up if the battery voltage is less than 2.5V (typical).

POWER-ON DEFAULTS

Upon application of power to the device, the following register bits are set to a '0' state: WDS; BMB0-BMB4; RB0-RB1; AFE; ABE; W; R; FT.

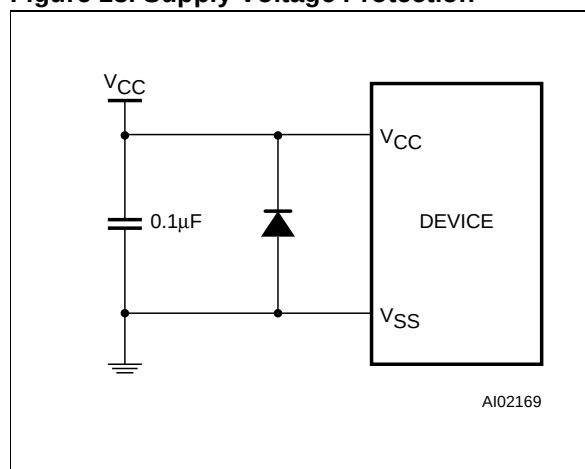
POWER SUPPLY DECOUPLING and UNDERSHOOT PROTECTION

I_{CC} transients, including those produced by output switching, can produce voltage fluctuations, resulting in spikes on the V_{CC} bus. These transients can be reduced if capacitors are used to store energy, which stabilizes the V_{CC} bus. The energy stored in the bypass capacitors will be released as low going spikes are generated or energy will be absorbed when overshoots occur. A ceramic bypass capacitor value of $0.1\mu\text{F}$ (as shown in Figure 14) is recommended in order to provide the needed filtering.

In addition to transients that are caused by normal SRAM operation, power cycling can generate neg-

ative voltage spikes on V_{CC} that drive it to values below V_{SS} by as much as one Volt. These negative spikes can cause data corruption in the SRAM while in battery backup mode. To protect from these voltage spikes, it is recommended to connect a schottky diode from V_{CC} to V_{SS} (cathode connected to V_{CC} , anode to V_{SS}). Schottky diode 1N5817 is recommended for through hole and MBRS120T3 is recommended for surface mount.

Figure 13. Supply Voltage Protection



M48T559Y

Table 13. Ordering Information Scheme

Example:	M48T559Y	MH	1	TR
Device Type M48T				
Supply Voltage and Write Protect Voltage 559Y = V _{CC} = 4.5V to 5.5V; V _{PFD} = 4.2V to 4.5V				
Package MH ⁽¹⁾ = SOH28				
Temperature Range 1 = 0 to 70 °C				
Shipping Method for SOIC blank = Tubes TR = Tape & Reel				

Note: 1. The SOIC package (SOH28) requires the battery/crystal package (SNAPHAT) which is ordered separately under the part number "M48T28-BR12SH1" in plastic tube or "M4T28-BR12SH1TR" in Tape & Reel form.

Caution: Do not place the SNAPHAT battery/crystal package "M4T28-BR12SH1" in conductive foam since will drain the lithium button-cell battery.

For a list of available options (Speed, Package, etc...) or for further information on any aspect of this device, please contact the ST Sales Office nearest to you.

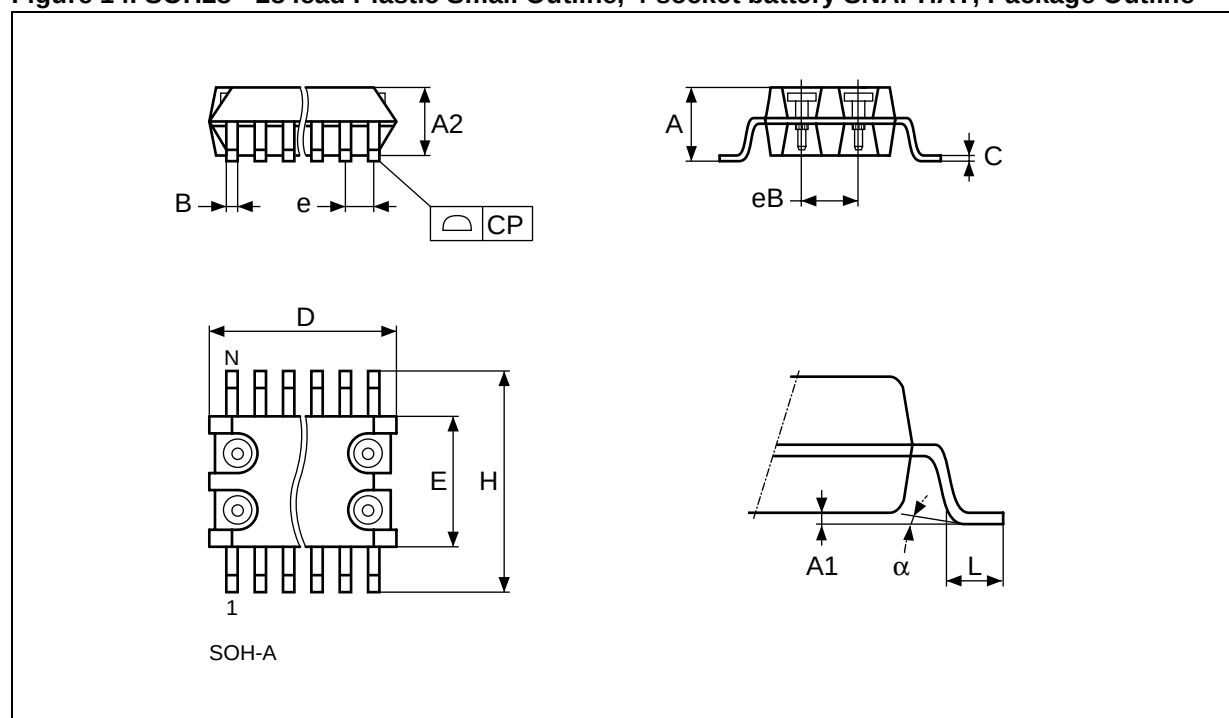
Table 14. Revision History

Date	Revision Details
June 1998	First Issue
02/07/00	Description Paragraph changed Setting Alarm Clock paragraph changed M4T28-BR12SH SNAPHAT Housing for 48mAh Battery & Crystal Package added (Table 16) Power Down/Up Mode AC Waveforms changed (Figure 5) Back-up Mode Alarm Waveforms changed (Figure 11)

Table 15. SOH28 - 28 lead Plastic Small Outline, 4-socket battery SNAPHAT, Package Mechanical Data

Symb	mm			inches		
	Typ	Min	Max	Typ	Min	Max
A			3.05			0.120
A1		0.05	0.36		0.002	0.014
A2		2.34	2.69		0.092	0.106
B		0.36	0.51		0.014	0.020
C		0.15	0.32		0.006	0.012
D		17.71	18.49		0.697	0.728
E		8.23	8.89		0.324	0.350
e	1.27	–	–	0.050	–	–
eB		3.20	3.61		0.126	0.142
H		11.51	12.70		0.453	0.500
L		0.41	1.27		0.016	0.050
α		0°	8°		0°	8°
N	28			28		
CP			0.10			0.004

Figure 14. SOH28 - 28 lead Plastic Small Outline, 4-socket battery SNAPHAT, Package Outline

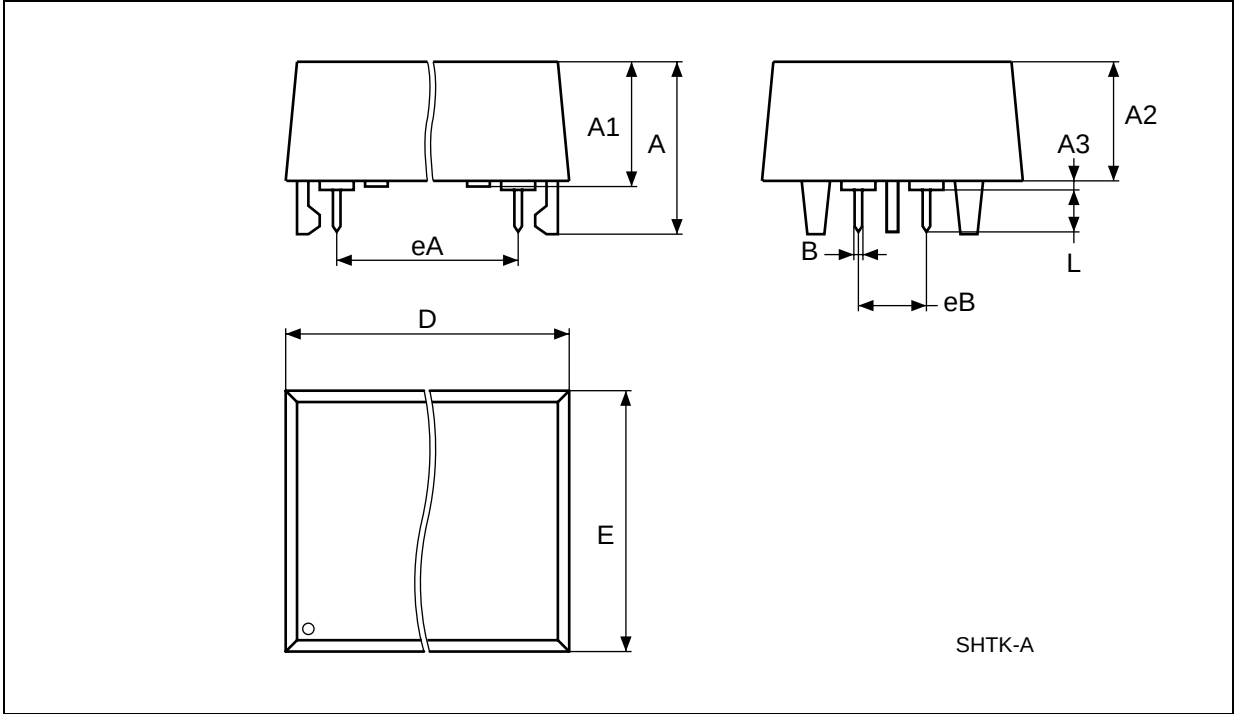


Drawing is not to scale.

Table 16. M4T28-BR12SH SNAPHAT Housing for 48mAh Battery & Crystal, Package Mechanical Data

Symb	mm			inches		
	Typ	Min	Max	Typ	Min	Max
A			9.78			0.385
A1		6.73	7.24		0.265	0.285
A2		6.48	6.99		0.255	0.275
A3			0.38			0.015
B		0.46	0.56		0.018	0.022
D		21.21	21.84		0.835	0.860
E		14.22	14.99		0.560	0.590
eA		15.55	15.95		0.612	0.628
eB		3.20	3.61		0.126	0.142
L		2.03	2.29		0.080	0.090

Figure 15. M4T28-BR12SH SNAPHAT Housing for 48mAh Battery & Crystal, Package Outline

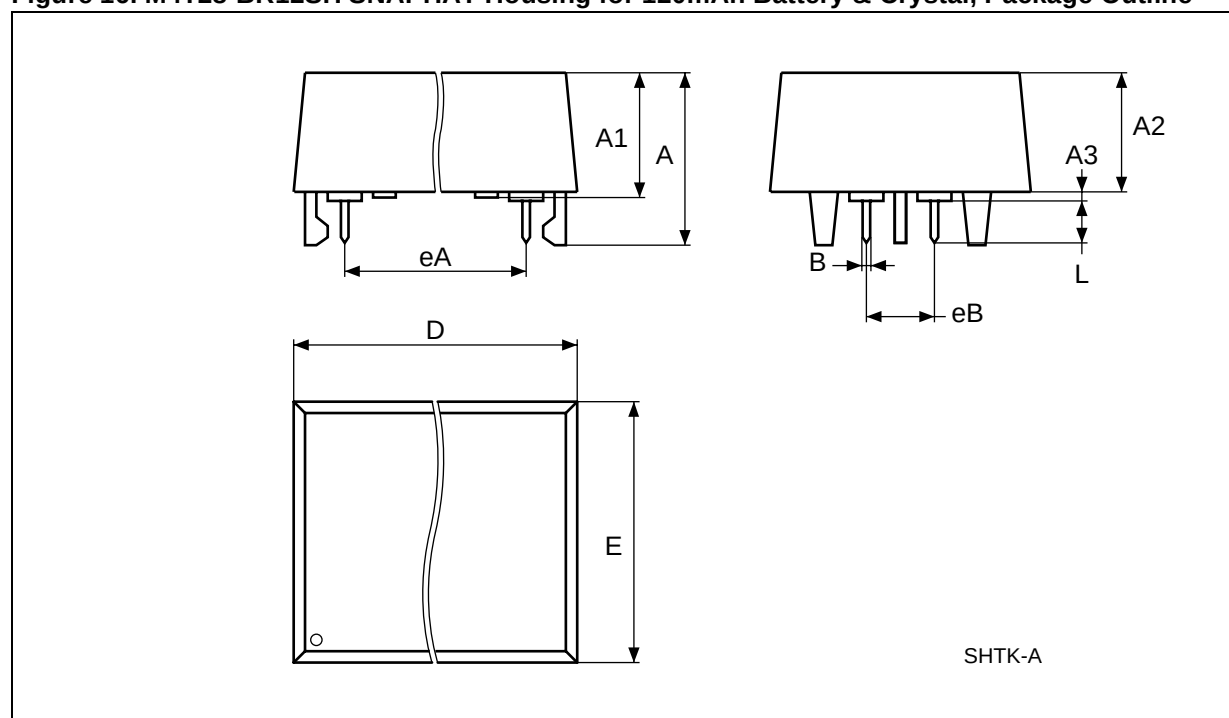


Drawing is not to scale.

Table 17. M4T28-BR12SH SNAPHAT Housing for 120mAh Battery & Crystal, Package Mechanical Data

Symb	mm			inches		
	Typ	Min	Max	Typ	Min	Max
A			10.54			0.415
A1		8.00	8.51		0.315	0.335
A2		7.24	8.00		0.285	0.315
A3			0.38			0.015
B		0.46	0.56		0.018	0.022
D		21.21	21.84		0.835	0.860
E		17.27	18.03		0.680	0.710
eA		15.55	15.95		0.612	0.628
eB		3.20	3.61		0.126	0.142
L		2.03	2.29		0.080	0.090

Figure 16. M4T28-BR12SH SNAPHAT Housing for 120mAh Battery & Crystal, Package Outline



Drawing is not to scale.

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